CDB/RDB

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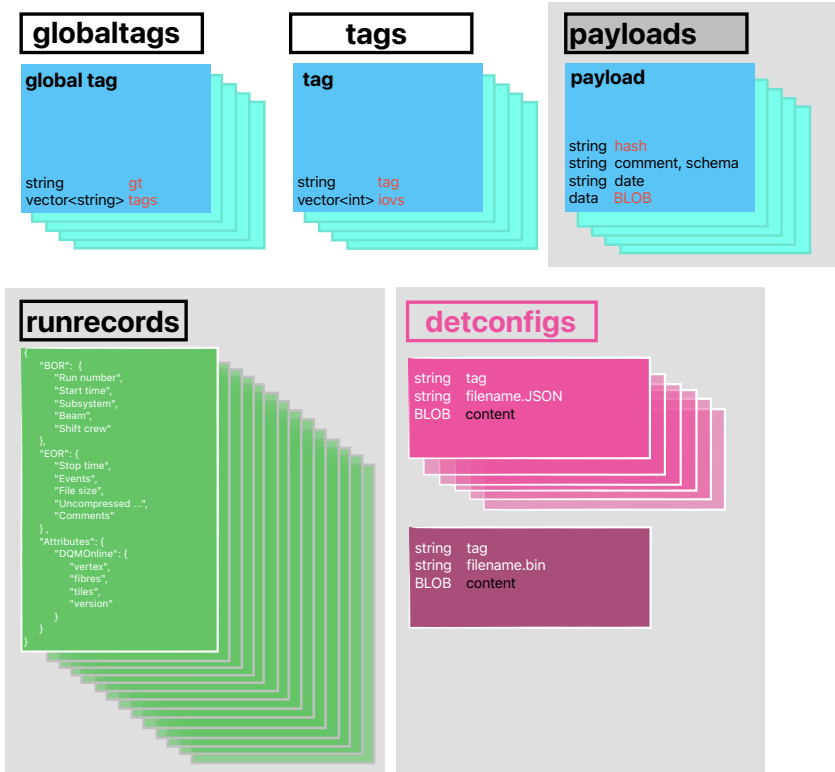
- CDB $\neq$ Conditions Database
- RDB $=$ Run Database

Introduction (why CDB ≠ Conditions DB)

- CDB has three "domains" (so far)
- RDB - Run Database
  - information about (all) Mu3e runs (time, duration, comments, DQM, . . .)
  - records can be inserted/updated via curl
  - web user interface (alternatively)
- Configuration data (for detectors)
  - can contain "anything"
    - mask/TDAC files
    - JSON dumps from ODB/MIDAS
    - unused so far - "unknown" VTX mask status
  - records inserted/extracted via curl
- Conditions for reco/vtx/ana (offline)
  - alignment constants,
    chip status, calibrations, . . .
  - retrievable in offline code/python/. . .



- Backends: REST api (http://mu3edb0.psi.ch via curl) and JSON files (local)
  - curl - "command line tool and library for transferring data with URLs" (works within PSI's network)

Why "Conditions"?

- Only way to do proper analysis on real detector data
  - every HEP experiment has one, often not very popular (myself included)
- Nothing is perfect forever
  - alignments change
  - chips (pixels) are bad, go bad, and/or recover
  - better calibrations become available and need to be applied
  - "conditions" change with runs
  - for every component in this experiment, not just "your" detector

⇒ you want/have to correct for imperfections "automatically"

→ you primarily want to focus on physics results, not corrections (right?)

⇒ CDB provides the tools.

→ Therefore you want one (if you think about it)

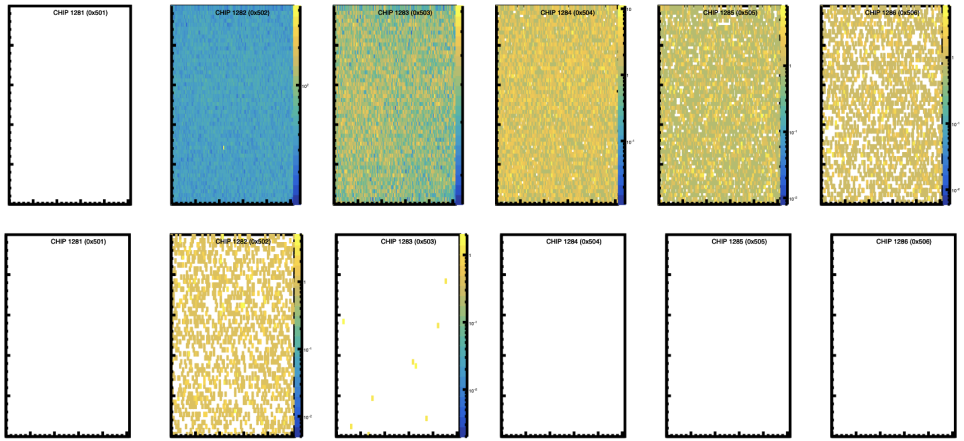
- A note on change frequency
  - runs are (by fiat Nik) the smallest granularity for detector changes
    TS run ranges (calEventStuffV1) are a violation of this rule
  - eras (= "large" run ranges) can be used for MC production

What is in the conditions DB?

- "Payloads" describe conditions data for analysis of detector data
  - ▷ payloads change if detector conditions change
  - ▷ different implications
    - noise (remove in data)
    - dead (remove in MC)
    - accomodate (in data and MC)
  - ▷ derived in "prompt" workflow (tomorrow)
  - ▷ derived in "offline"

Run 6157:

Run 6164:

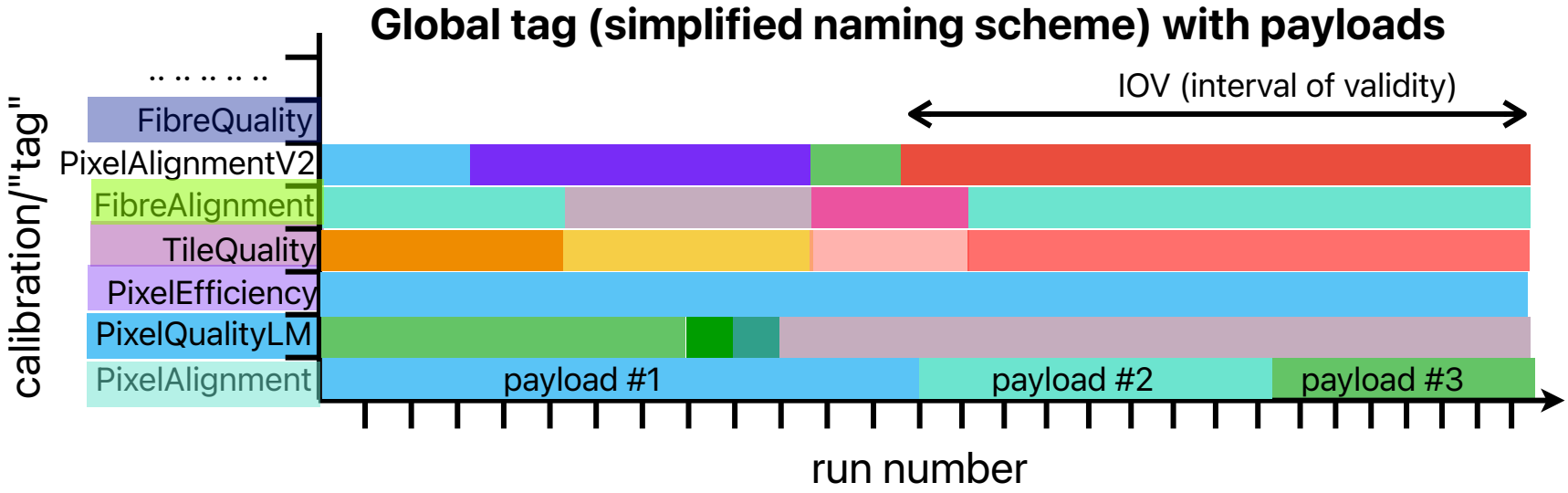


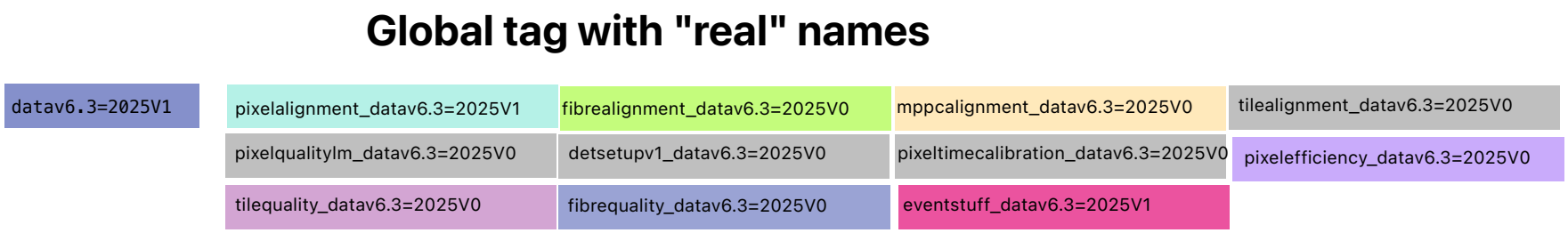
- Payloads belong to certain "calibrations" (e.g. pixelQualityLM)
  - ▷ they come in different versions
    - improvements (include MIDAS meta data)
    - bugfixes (include MIDAS meta data correctly)
    - more alignment analyses
  - ▷ they apply to certain run ranges (iov = interval of validity)
    - can change with versions

```
[moor>cat ~/data/mu3e/cdb/tags/detsetupv1_datav6.3=2025V0
  { "tag" : "detsetupv1_datav6.3=2025V0", "iovs" : [ 1, 2177, 6302 ] }
```

  - → there are (nearly) always multiple payloads for a calibration
    - "vector" of iovs

Global tags, tags, IOVs, payloads



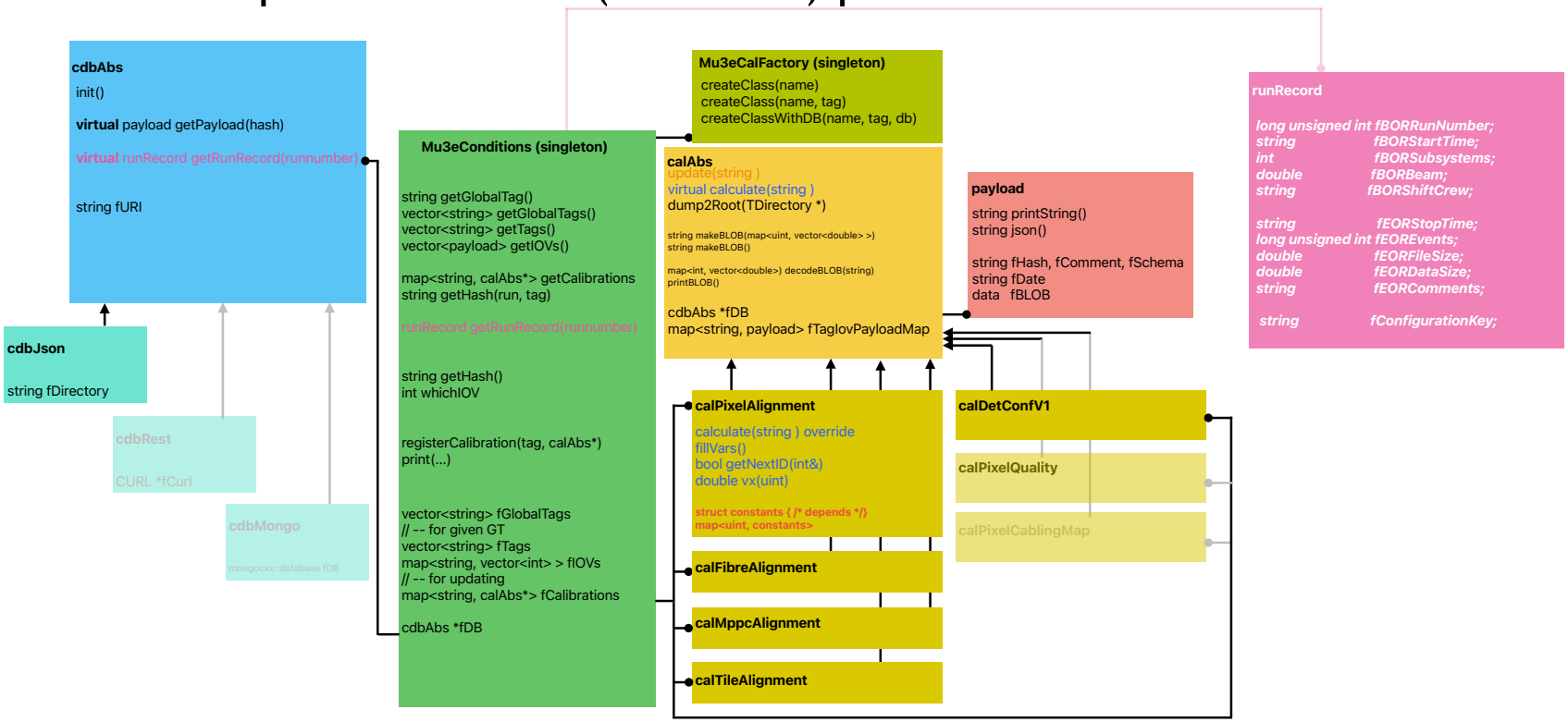


- Terminology
  - payload - conditions data describing detector status (conditions)
  - IOV - interval of validity (open ended: only start provided)
  - tag - name identifying "calibration", together with vector of IOVs
  - global tag (GT) - complete set of tags

Software architecture

code on github

- One central class (Mu3eConditions) to gain conditions access
- "Everything" coded in terms of abstract classes
  - ▷ DB backend can be replaced w/o problems
  - ▷ Constant specifics in one (concrete) place



CDB access Conditions management Calibration classes

CDB technicalities

- How-to (showing only CDB arguments)
  - ▷ connect to a JSON CDB and choose a GT

    mu3eTrirec --cdb.dbconn=/Users/ursl/data/mu3e/cdb --cdb.globalTag=mcidealv6.5=2025V0
  - ▷ test a different alignment (keeping rest of GT)

    mu3eTrirec --cdb.replaceTags=pixelalignment_datav6.3=2025V1test
  - ▷ replace multiple tags

    mu3eTrirec --cdb.replaceTags=pixelalignment_datav6.3=2025V1test,tilealignment_datav6.3=2025V0test
  - ▷ information about tags and GTs with command-line tool (will come in v6.5)

```
GT mcidealv6.5: MC ideal (=complete) detector geometry v6.5. No deficiencies, all 100% efficient.
  tag pixelalignment_mcidealv6.5: Ideal detector geometry with pixel alignment with MC truth from v6.5. Created for
  tag fibrealignment_mcidealv6.5: Ideal detector geometry with fibre alignment with MC truth from v6.5. Created for
  tag tilealignment_mcidealv6.5: Ideal detector geometry with tile alignment with MC truth from v6.5. Created for GT
  tag mppcalignment_mcidealv6.5: Ideal detector geometry with mppc alignment with MC truth from v6.5. Created for GT
  tag pixelqualitylm_ideal: Perfect pixel detector with no deficiencies. Created for GT=ideal
  tag fibrequality_ideal: Perfect fibre detector with no deficiencies. Created for GT=ideal
  tag tilequality_ideal: Perfect tile detector with no deficiencies. Created for GT=ideal
  tag pixelefficiency_ideal: Fully efficient for complete pixel detector. Created for GT=ideal
  tag eventstuffv1_ideal: No limitations to time stamps. Created for GT=ideal
  tag detsetupv1_mcidealv6.5: Created for GT=mcidealv6.5
```

- Caveat: For a given run, the CDB will give you what it should!
  - ▷ e.g. if you mu3eSort MC data for run 4602
    - you'll have many bad links
    - you'll have very low pixel "efficiency"
  - → this can only be fixed with different (better) payloads

The current global tags

- Quite a mess and reflecting the push to the BVR
  - tags created before payloads available
  - payloads replaced
  - chaos
- datav6.3=2025V0
  - corrections for MC for BVR
    pixel alignment(?) and qualities (px bad), valid TS ranges
- datav6.3=2025V1
  - corrections for data for BVR
    alignment with cosmics+metrology
- datav6.5=2025V0test
  - good pixel quality with pixelalignment_datav6.3=2025V1
- mcidealv6.5=2025V0test
  - good pixel quality with pixelalignment_mcidealv6.1=2025CosmicsVtxOnly
- mcidealv6.5
  - "bootstrap" GT generated during make install with complete detector

CDB/RDB size matters

- mu3edb0: new (2U) database server in counting house
  - ▷ 128GB DDR4 RAM, 8 Core Intel XEON Silver 4309Y 2.8GHz Processor
  - ▷ 2x24TB disk space (RAID'ed to 24 TB), currently
- In 2025 mongodb backend
  - ▷ RDB: 2.6MB runrecords
  - ▷ CDB: 250MB payloads
    12GB attachments (pdf summaries)
- JSON backend (for rsync)

```
login001>du -ks /data/project/mu3e/cdb
1344880 /data/project/mu3e/cdb
```

- JSON backend no longer limited by too many files
  - ▷ CDB: payloads in run blocks for each tag
  - ▷ RDB: run records in run blocks
  - → no more hesitation to keep tags/payloads/etc

Conclusions

- Technical improvements for v6.5
  - ▷ Comments in GT and tags for interactive documentation
  - ▷ JSON backend no longer file number limited
    - functionally fully equivalent to mongodb backend, finally
- RDB not really discussed today
  - ▷ it works (if you think otherwise you should have told me!)
  - ▷ future improvements
    - add more fields (e.g., TS ranges, for events and subsystems), API additions
    - full mirror to JSON backend (minus attachments?)
    - technical bugfixes (for deleted attachments, . . .)
- CDB works reasonably well technically
  - ▷ GT and tag management needs to be improved (I need tools)
    (note: payload preparation to be discussed tomorrow, not CDB proper)
    MC should be prepared in eras (subsystem definitions)
  - ▷ h/w should survive 2026 data taking (and likely also 2027)
    (many empty HDD slots available)
  - ▷ add configuration data (e.g. pixel masks)
    also for offline payload production! (i.e. not on mu3ebe)

CDB Discussion

- What global tags are required?
  - ▷ "Data"
    - remove noisy pixel, LVDS error links, similar for fibres/tiles, . . .
    - apply improved alignment information to geometry
    - magnet status
    - provide calibration?? (pixeltimecalibration)
  - ▷ "MCideal"
    - perfect detector (possibly different configurations)
    - can be modified with dedicated tag(s) for studies
  - ▷ "MCrealistic"
    - existent detector with deficiencies (defects/noise/resolution)
    - not necessarily with the same iovs as data (better: eras)
  - ▷ "analysis"
    - For "later" stages (both in stages of analysis and wall-clock time)
    - pixelEfficiency, (pixel)TimeCalibration, . . .
- why this effort?
  - → to have identical code process detector and MC data (important)

Discussion

- Include (pixel) mask information?
  - ▷ can be incorporated into pixelQualityLM
  - ▷ but need to be available for prompt workflow
    - unambiguous, reproducible
    - and also for re-processing of prompt workflow
- Use-cases for CDB (who uses what when how often at which stage)
  - ▷ MIDAS meta data move out of CDB into RDB? (remove completely?)
- What to do about non-usage of CDB in certain cases
  - ▷ B field in trirec?
- should analysis tags be part of a "reco" GT?
- (Global) tag management not trivial
  - ▷ naming scheme likely to change
    e.g. pixelalignment_datav6.3=2025V1 $\rightarrow$ pixelalignment_cosmics=metrology=V1?
  - ▷ cutting corners not helpful
  - ▷ once again: apologies!